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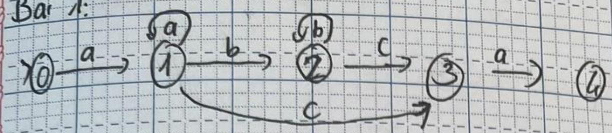
21CE067

Bài tập DFA
Automat và ngôn ngữ hình thức

DFA

Bài 1:



a) $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{0, 1, 2, 3, 4\}$

$\Sigma = \{a, b, c\}$

$q_0 = 0$

$F = \{4\}$

δ : $\delta(0, a) = 1$; $\delta(1, a) = 1$, $\delta(1, b) = 2$

$\delta(1, c) = 3$; $\delta(2, b) = 2$; $\delta(2, c) = 3$

$\delta(3, a) = 4$

b) X: ~~aa, bbc~~ aabbca , abbbca , abbaca, aaaca

0aabbca	0abbbca	0abbaca	0aaaca
⇒ 1abbca	⇒ 1bbbca	⇒ 1bbaca	⇒ 1aaca
⇒ 1bbca	⇒ 2bbca	⇒ 2bbaca	⇒ 1aca
⇒ 2bca	⇒ 2bca	⇒ 2baca	⇒ 1ca
⇒ 2ca	⇒ 2ca	⇒ 2aca	⇒ 3a

$\Rightarrow 3a$ $\Rightarrow 3a$ $\times$ $\Rightarrow 4 \in F$

$\Rightarrow 4 \in P$ $\Rightarrow 4 \in F$

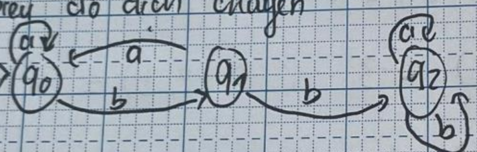
Vậy xâu aabbca, abbbca, aaaca là được đón nhận

Bài 2:

Bảng dịch chuyển

δ	a	b
q_0	q_0	q_1
q_1	q_0	q_2
q_2	q_2	q_2

Biểu đồ dịch chuyển



b) Tìm $L(M)$

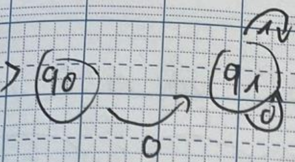
$L(M) = \{ a^n b (ab)^m b a^n b^m$ với $n, m \geq 0 \}$

Bài 3

a) $L(M) = \{ 0(10)^n$ với $n > 0 \}$

$M_a = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_1\})$

δ: $\delta(q_0, 0) = q_1$, $\delta(q_1, 0) = q_1$, $\delta(q_1, 1) = q_1$

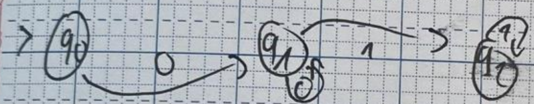


c) $L(M) = \{0^n 1^m$ với $n \geq 0, m > 0\}$

$M_c = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_1\})$

δ: $\delta(q_0, 0) = q_0$, $\delta(q_0, 1) = q_1$, $\delta(q_1, 1) = q_1$

b) $L(M) = \{0^m 1^m$ với $m > 0, m > 0\}$

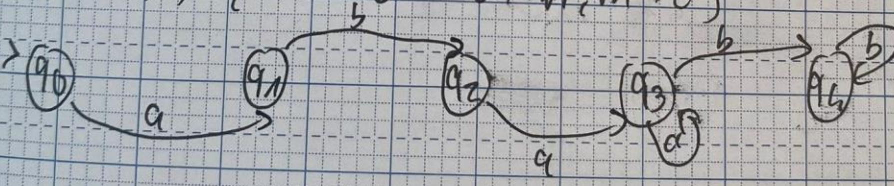


$M_b = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$

δ: $\delta(q_0, 0) = q_1$, $\delta(q_1, 0) = q_1$, $\delta(q_1, 1) = q_2$

$\delta(q_2, 1) = q_2$

d) $L(M) = \{a b a^n b^m$ với $n, m > 0\}$

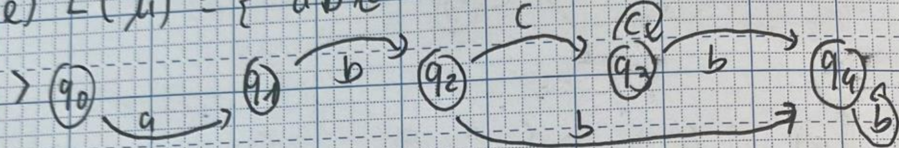


$M_d = (\{q_0, q_1, q_2, q_3, q_4\}, \{a, b\}, \delta, q_0, \{q_4\})$

δ: $\delta(q_0, a) = q_1$, $\delta(q_1, b) = q_2$, $\delta(q_2, a) = q_3$

$\delta(q_3, a) = q_3$, $\delta(q_3, b) = q_4$, $\delta(q_4, b) = q_4$

e) $L(M) = \{ abc^n b^m$ với $m \geq 0, m > 0 \}$



$M_e = (\{q_0, q_1, q_2, q_3, q_4\}, \{a, b, c\}, \delta, q_0, \{q_4\})$

δ: $\delta(q_0, a) = q_1$, $\delta(q_1, b) = q_2$, $\delta(q_2, c) = q_3$

$\delta(q_2, b) = q_4$, $\delta(q_3, c) = q_3$, $\delta(q_3, b) = q_4$

$\delta(q_4, b) = q_4$.